

AUTOMATED OPTIMAL CAMERA PLACEMENT FOR MARKERLESS BIOMECHANICS: A COMPARATIVE STUDY AGAINST MANUAL SETUPS

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Introduction

The accuracy of markerless motion capture (MMC) is fundamentally constrained by 3D body coverage and multi-view redundancy. Research demonstrates that suboptimal camera placement induces joint angle errors of up to 20°, even in restricted capture volumes such as hand tracking [1]. In full-body biomechanics, insufficient spatial distribution severely reduces multi-view keypoint visibility and triangulation robustness, particularly during high-speed or partially occluded actions [2]. Despite these critical dependencies, laboratory camera placement remains largely manual and empirical. Relying on expert intuition in complex environments often results in undetected blind spots and suboptimal bilateral coverage. To address this methodological gap, we developed the Lab Camera Optimizer, an open-source computational tool designed to systematically maximize 3D line-of-sight coverage. This pilot study evaluates the algorithm's efficacy by comparing an existing expert-designed manual setup against an automated, optimized configuration within an active biomechanics laboratory.

<https://github.com/flodelaplace/lab-camera-optimizer>

Methods

The Lab Camera Optimizer evaluates candidate layouts within a configurable 3D space, incorporating room geometry, occluding obstacles, and camera constraints (FOV, range, mounting heights). The algorithm computes a global composite score across a sample grid by integrating four key variables: strict vertical body coverage ($\geq 90\%$), inverse-square distance attenuation, angular diversity to penalize redundant viewpoints, and a bilateral factor. This biomechanically critical factor uses a geometric mean of scores from opposing sides of the movement axis to enforce balanced 3D reconstruction.

To navigate this combinatorial space, the optimizer employs a randomized multi-start algorithm combining a farthest-point sampling initialization with a Greedy 1-opt local search. For this pilot study, we bypassed the tool's ability to co-optimize the capture zone location. Instead, we fixed the zone to its current physical position to ensure a direct comparison between the existing expert-designed manual setup and the new automated configuration, using the identical nine-camera mixed inventory for both.

Results

The automated configuration increased the global coverage score by 58% (576.0 vs. 363.8) compared to the manual setup. Crucially, the optimized layout

achieved the target spatial redundancy of ≥ 6 cameras across 100% of the primary analysis zone, a stark contrast to the manual setup which reached this target in only 28% of the zone. This gain was primarily driven by the algorithm's utilization of mixed camera orientations (three portrait, four landscape) to maximize vertical overlap, whereas the manual setup exclusively employed landscape views. Furthermore, a combinatorial sweep evaluating six potential capture zone locations confirmed that the experts' manually chosen placement was indeed the mathematically optimal sub-zone within the room.

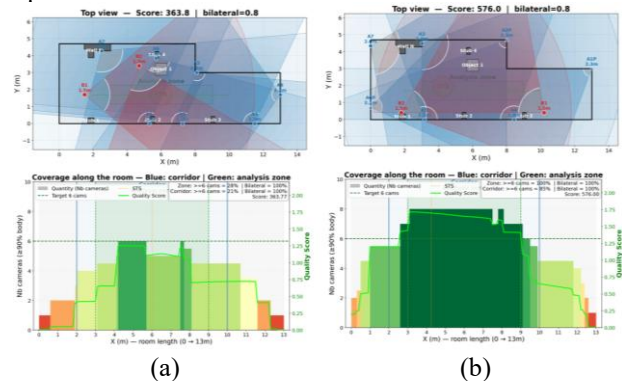


Figure 1: Visual representation of the manual (a) and optimized (b) camera setups. Top row displays the top-down room layouts, obstacles, and camera fields of view. Bottom row illustrates the coverage along the analysis zone with the number of overlapping cameras (bar chart) and the local quality score (green line).

Discussion

These findings highlight a crucial dichotomy in laboratory design: while expert intuition excels at macro-level experimental planning—perfectly identifying the optimal movement zone—it struggles to resolve the high-dimensional spatial puzzle of multi-camera placement. By objectively optimizing parameters such as portrait/landscape combinations and strict line-of-sight constraints, the *Lab Camera Optimizer* drastically improves 3D coverage and point redundancy without requiring additional hardware investments. Although this pilot study is limited to a comparison against a single manual setup, this open-source tool bridges the gap between empirical setups and computational optimization, providing a systematic approach to standardize markerless environments and secure high-fidelity kinematic data collection.

References

1. Mulla et al, J Biomech, 181:112534, 2025.
2. Li et al, Sensors, 26:1689, 2026.